

Note No :1



Ref:SMTTP/ENVT. MGT./2023-24/EMG-AU/546341

Date:06/10/2023(17:23:14)

Subject:Approval of LMI- The Control of environmental pollution Rev 06

The following LMI pertaining to Simhadri-EMG is submitted hereby for approval,

LMI/OGN/ENV/001 Rev No 06- THE CONTROL OF ENVIRONMENTAL POLLUTION

Revision of OGN is envisaged in this revision in line with COS-ISO-00-OGN/ENV/001 Rev No3, proposed for approval of revision.

The details of revision are as follows,

- 1.Section -4 (4.1) has been revised for inclusion of FC. (Page No.04)
- 2.Section-4 Water cess act applicability omitted. (Page No.04)
- 3.Section-4: Inclusion of section 4.3.7 for waste management.(Page No.05)
- 4.Section-5 Waste management policy included.(Page No. 06)
- 5.Section-6: Minor changes. S.No. 6.3 (Page No.07)
- 6.Section -8 -8.3.2 & 8.3.3 Chief general manager changed to HoP Simhadri(Page No.09)
- 7.Section-9 Omission of Water cess act.(Page No.11)
- 8.Section-10 Frequency of revision of document change for two years as per OS circular.(Page No.12)

Revised LMI is enclosed for reference.

Submitted for approval please.

**Dr Pradeep M
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Submitted by:

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Note No:2

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Pl review.

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Note No:4

LMI has been reviewed and found in order w.r.t equipment availability, calibration of equipment and Schedule of Environmental parameters monitoring which are already in practice as per the LMI.

May please forward.

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Note No:6

Reviewed and put up for further approval.

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Note No:7

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Note No:11

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LOCATION MANAGEMENT SYSTEM

LMI/OGN/ENV/001

Rev. No:06

THE CONTROL OF ENVIRONMENTAL POLLUTION

Approved for Implementation by:HOP

(SIMHADRI)

Date :06.10.2023

Enquiries to EMG Dept.

LMI APPROVAL DOCUMENT

1. TITLE OF LMI	THE CONTROL OF ENVIRONMENTAL POLLUTION
2. a) DOCUMENT CODE b) REVISION/ISSUE NO. & DATE	LMI/OGN/ENV/001 REV.6 / 06.10.2023
3. ORIGINAL/REF. DOCUMENT	COS-ISO-00-OGN/ENV/001 Rev. No.: 1 Date: 29.11.2018
4. REVISION MADE BY	Dr Pradeep M, Manager -EMG
5. CHECKED & REVIEWED BY	Dr V Jayan, AGM (HoD -EMG &AU)
6. REVIEWED BY	HOD (CHEM)
	HOD (C&I)
	GM (Operations)
	GM (O&M)
7. DATE OF ISSUE	06.10.2023
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THE CONTROL OF ENVIRONMENTAL POLLUTION

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THE CONTROL OF ENVIRONMENTAL POLLUTION

1.0 SUMMARY

1.1 SUPERSEDED DOCUMENTS

LMI/OGN/ENV/001 Rev No 05

1.2 Scope: The control of Environment Pollution

1.3 Review:

Review responsibility - HOD (EMG)

Review frequency – 2 Yearly basis, or as and when required, whichever is lower

2.0 INTRODUCTION

This Location Management Instruction describes NTPC-Simhadri's approach in managing environmental impact due to discharge of solid, liquid and gaseous substances to the environment, and handling wastes and by products. It explains the Regulatory and Policy framework, defines and gives guidance on how the control of emissions and discharges shall be affected, monitored and reported within the station and describes the way NTPC, Simhadri will interface with the Enforcing Authorities.

3.0 REVISION DETAILS

COS-ISO-00-OGN/ENV/001 Rev No 03

4.0 LEGAL AND REGULATORY FRAMEWORK (ENVIRONMENT)

4.1 The construction and operation of power stations is governed by requirements to comply with certain conditions stipulated in Environmental Clearance, NOC (NO OBJECTION CERTIFICATE), Consent to establish, Forest Clearance if applicable and consent to operate stipulated for the plant/ station in actual or deemed and statutory provisions for controlling emissions to air and water and for handling and disposing of solid waste/Hazardous waste/ e-Waste and Bio-medical waste. To operate the power station, it is necessary to have the Consent Order From A.P. State Pollution Control Board (APPCB) and to be renewed as per the requirement. During project clearance stage No Objection Certificate (NOC) and Environmental Clearance (EC) are issued by AP State and Ministry of Environment & Forest, Govt. of India, respectively.

4.2 To operate a power station it is mandatory to have the Consent Order from respective State Pollution Control Board (SPCB)/ POLLUTION CONTROL COMMITTEE/ENVIRONMENT CONSERVATION BOARD. During project clearance stage No Objection Certificate (NOC) and Environmental Clearance (EC) are issued by concerned State and Ministry of Environment, Forest and Climate Change, Govt. of India, respectively. Moreover, the various rules and regulations which are framed, updated and amended by the MoEF &CC, CPCB and state regulators are also to be complied

4.3 During the last decade there have been major changes in the Regulations

controlling pollution from Power Stations stemming mainly from various Pollution Control Acts, Rules & Notifications issued there under. The main impact on NTPC Simhadri are:

- 4.3.1 Power generation processes are regulated under The Factory Act 1948, Water (P&CP) Act, 1974 and amendments thereof. The Forest Conservation Act 1980 (Amended 1988), The Indian Forest (Amendment) Bill, 2012, Forest (Conservation) Amendment Rules, 2014, The Air (P&CP) Act 1981 and amendments thereof, The Environment Act 1986, The Public Liability Insurance Act, 1991 and various amendments, notifications, guideline and rules issued under these Acts and outcomes of various petitions in NGT.
- 4.3.2 Simhadri Power Station can operate under the Consent Order, granted by SPCB and in accordance with the conditions detailed in that Consent Order. Even where not expressly stated as a condition of Consent Order, there is implied in it that the person carrying on the process must use the best available technique.
- 4.3.3 For preventing the release of prescribed substance or, where that is not practicable for reducing to a minimum the release of such substances and for rendering harmless of any such substances which are so released.
 - For rendering harmless any other substances which might cause harm if released into any environmental medium.
- 4.3.4 SPCB renews Consent Order from time to time and may specify variations to the Consent Conditions.
- 4.3.5 The Consent Order includes limits on the release of any substance.
- 4.3.6 Emissions to atmosphere, discharges to water bodies and solid waste management will be regulated by various conditions laid in the Consent Order and authorization accorded by SPCB/ PCC under BMW (M&H) Rule, HW (M&HTB) Rule, EEW (M&H) Rule, Battery (Management& Handling) Rule. There may also be certain conditions in this regard in NOC/EC.
- 4.3.7 The Transportation, handling, storage and disposal of Municipal solid waste, Bio Medical Waste, Hazardous waste, E waste, Plastic waste, EPR shall be governed by respective Acts/rule or conditions stipulated in EC/CTO/Authorizations and guidelines by CPCB/SPCB.
- 4.3.8 Controls on the transport and disposal of waste solids are also governed under various conditions being laid now in MoEF notifications & amendments thereof and consent orders, especially with regard to disposal, transportation & utilization of ash. Conditions in this regard for ongoing projects/stations or future projects may also appear in NOC/EC.

5.0 NTPC ENVIRONMENT POLICY

One of the NTPC Corporate objectives is "To provide cleaner energy by committing to highest possible levels of performance in environmental compliance, practices and stewardship."

NTPC takes environmental care as a guiding principle and the company has developed its Environment Policy, Water Policy, Rainwater harvesting Policy, Biodiversity Policy, Plastic policy and **waste Management Policy implementation.**

6.0 IMPLEMENTATION

6.1 The Simhadri has to aim implementing the following programmes:

6.1.1 At the planning stage, by identification of environmental constraints and opportunities and the development of cost-effective solutions.

6.1.2 At the design and construction stage, by identification of sensitive environmental impacts and issues and their reduction to acceptable levels.

6.1.3 At the operational stage, by control of environmental impacts through the cost effective use of technology, operating practice and conservation.

6.2 Within O&M, a system has to be established for environmental measurement and monitoring. The station has obtained ISO 14001 certification for environment management system which consists of auditable systems to comply with environment programmes and all statutory requirements to control pollution.

Keeping in view of the above

6.2.1 The Company's Environmental Programme, Codes of Practice and Guidelines are converted in to site specific targets and practices as per Annexure A

6.2.2 Training and awareness programmes are to be conducted to make staff work on the site aware of their environmental responsibilities.

6.2.3 A file of environmental obligations and an inventory of emissions, discharges and wastes is to be maintained.

6.2.4 The environmental parameters are to be monitored as per schedule to assess the environmental impacts of the station.

6.2.5 Within the site operating and development plan, environmental improvement proposals covering plant performance etc are to be considered.

6.2.6 Operate a quality assured system for monitoring all significant releases and the local environment.

6.2.7 Station Head is apprised of the environmental performance against all regulatory, public and Company requirements regularly at discussions at appropriate forums.

6.2.8 A local environmental management system is in place identifying staff responsibilities, site standards, limits and management instructions as per Annexure A. The above will help in identifying and managing all environmental risks to minimize liability from

environmental incidents.

6.2.9 A disaster management plan is in place to take care off, all-environmental accidents. Non-compliance incidents and public complaints are to bereviewed.

6.2.10 Ensure that all environmental accidents, non-compliance incidents and public complaints are investigated and handled in a manner to minimise loss of the Company's reputation

6.3 Simhadri station will also assist NTPC to maintain its position as a knowledgeable buyer and operator of Plant and encourage suppliers to develop improved technology, diagnostics, control techniques and operational methods.

7.PLANNING, MONITORING AND COMPLIANCE

The over-all responsibility for implementing these requirements and meeting Corporate and Regulatory standards on operational plant/sites resides with the Head of the Station.

8.0 DETAILED REQUIREMENTS

8.1 Obligations - Definition of Standards and Targets

8.1.1 All standards, targets, limits or other conditions (e.g. condition of consent, local agreements if any, law regulatory instruments, like NOC/EC etc. company policy standards or targets), which regulate environmental performance or be deemed to be a condition of operating the plant/process.

8.1.2 Any inventory of these obligations are updated by EMG regularly

8.2 Emissions Inventory

8.2.1 A continual assessment and record of the nature and quantities of emissions and discharges produced at the NTPC Simhadri (by NTPC or other agencies Engaged by NTPC) and of waste and by-products imported to, produced at, stored at or dispersed from the station are to be made. The detailed requirements are:

8.2.1.1 Inventories to be complied and updated on a regular basis.

8.2.1.2 Annual records are compiled by 30th June and environment statement to be submitted to SPCB before 30th Sept.

8.2.1.3 Content include:

8.2.1.3.1 CO₂ Emission

8.2.1.3.2 SO₂ Emission

8.2.1.3.3 NO_x Emission

8.2.1.3.4 Particulate Emissions

8.2.1.3.5 Mercury Emissions

For (a) to (d) give Unit wise annual average, emission rate (mg/Nm³) and total emission (Tonnes/Year) are given

8.2.1.3.6 Ash Sales and Disposals (Qty)

8.2.1.3.7 Liquid Discharges (Qty, each category wise)

8.2.1.3.8 Feed stock Quality, Quantity and Origin

8.2.1.3.9 Waste Quality, Quantity and Destination (category wise)

8.2.1.3.10 Releases of Scheduled or hazardous substances (Substance wise), if any

8.2.1.3.11 Contamination of land or site waters if any

8.2.1.3.12 Environment related Public Complaints if any

8.2.1.3.13 Record of all types of waste inventories their storage locations / disposal details as per the provisions in the existing acts/Rules/ guidelines and conditions in EC/NOC/CTO/ concerned authorizations or any other direction/ notice from regulator/ court/NGT.

8.3 Assessment of Compliance.

8.3.1 The O&M in-charge must ensure that equipment installed for the control and monitoring of pollution is used, operated and maintained within its design intent.

8.3.2 The HOD (EMG), the delegate of **HoP Simhadri** is responsible for ensuring and demonstrating to the Regulatory Enforcing Authorities that the operation of his plant complies with authorized standards, targets and limits. He shall give the Enforcing Authority assistance to assess compliance and investigate incidents and complaints.

8.3.3 The HOD (EMG), the delegate of **HoP Simhadri** should seek to agree the appropriate indicators and remedial procedures with the local inspector of the appropriate Enforcing Authority. The levels set should represent values readily achievable by the plant over its full range of authorized feed stocks and loading, when maintained in good condition and when being operated in manner appropriate to its design intent, excluding known malfunction.

1. NTPC Simhadri being ISO 14001 shall take necessary actions necessary actions to comply all the audit points pertaining to surveillance / certification audits and internal audits by CC-EMG

2. Necessary programmes are to be taken up for continual improvement in environmental conditions.

8.3.4 Local operating instructions are produced to advise staff of the requirements for meeting agreed targets, limits and to give guidance on the operational, diagnostic and maintenance practices to be adopted. These instructions also specify mandatory instrumentation, record needs. Suitable corrective action

is taken when non-compliances with targets/limits occur or instruments fail.

8.3.5 Operational records are maintained to demonstrate historic compliance with requirements, to facilitate performance review, and to help resolve complaints.

8.3.6 Ability to meet current/expected limits and prevent recurrence of previous problems on this or similar sites are considered when formulating plant maintenance schedules. This also can be considered for refurbishment plans or enhancement proposals.

8.4 Need for plant Improvement

8.4.1 The EMG will consider and initiate proposal for refurbishment of pollution control (or related) equipments wherever there is difficulty in sustaining the required level of performances using normal repair/maintenance techniques.

8.4.2 The EMG will consider Plant improvement whenever:

8.4.2.1 There are pollution problems, incidents, complaints or nuisance.

8.4.2.2 There is difficulty in sustaining Authorized limits, or Company targets between maintenance periods despite the plant being operated in the prescribed manner.

8.4.2.3 Operational or feed stock changes are planned or necessitated which take the existing pollution control plant outside its design range or emissions outside authorized limits.

8.4.3 The Enforcing Authority may also request or require plant improvements.

This may be on one of the following grounds

8.4.3.1 There has been a change of law.

8.4.3.2 A current limit no longer applies for that plant.

8.4.3.3 An opportunity has arisen which the Officer-in-charge believes will enable plant performance improvement conveniently and economically (e.g. a substantial change is proposal)

8.5 Improvement Planning

8.5.1 All suggestions, proposals for major plant improvement activities will be

put up to Corporate Engineering for any modification plan.

8.6 The Interface with the Regulatory Authority

8.6.1 The implementation of NTPC pollution control programme involves a continual dialogue with the Enforcing Authority. This shall be structured as follows:

8.6.1.1 The EMG department will maintain the local interface with the Enforcing Authorities at Regional level. This shall be concerned with the day to day control of pollution at the site and compliance with its Regulatory requirements.

8.6.1.2 For the issue coming above the preview of Regional Office and the regulatory authority, the dialogue is to be made with zonal office, task force department and Head Office of State Pollution Control Board as per the requirement.

8.7 Reporting

8.7.1 A common reporting schedule is given by Corporate Environment Management Group to ensure that essential Corporate Requirements through Parivesh is to be met with.

8.7.2 An Environmental Complaints or incidents entailing an actual or alleged breach of compliance or nuisance claim, or any other item likely to go public, shall be reported immediately with action plan to Regional Head Quarter and also to Corporate Environment Management Group who in turn will report to other concerned.

9 REFERENCE DOCUMENTATION

1. Environment (Protection) Amendment Rules 2015
2. E-waste (Management & handling) Rules 2011
3. Notification on Ash Utilisation, 1999, 2003 and 2009
4. Water(Prevention & control of pollution)Act,1974
5. Water (Prevention & control of pollution) rule,1975
6. Air (Prevention & control of pollution) Act,1981
7. Air (Prevention & control of pollution) Rule,1982
8. The Environment (Protection)Act,1986
9. The Environment (Protection) Rules,1986
10. The Batteries(Management & Handling) Rules,2001
11. Plastic Waste Management Rules 2016
12. e-waste (Management) Rules, 2016

13. Bio-Medical Waste Management Rules,2016
14. Construction and Demolition Waste Management Rules, 2016
15. Hazardous and Other Wastes (Management and Trans boundary Movement) Rules, 2016
16. Solid Waste Management Rules, 2016

10. REVIEW

HOD (EMG) shall be responsible for reviewing of this document once in every two years or as necessary

Annexure- A
Responsibility Centers

S.No.	Activity	Responsibility
3	LEGAL AND REGULATORY FRAMEWORK ENVIRONMENT Monitoring , compliance and Renewal of all statutory compliances as detailed under 3.1, 3.2 3.3 as detailed in LMI	HOD EMG
4	NTPC ENVIRONMENT POLICY Compliance to NTPC Environment policy	HOD EMG
5	IMPLEMENTATION Monitoring of implementation of the guidelines as detailed under 5.1 & 5.2 as per NTPC environmental policy and clause 6 of LMI	HOD EMG
7	DETAILED REQUIREMENTS	
	7.1 Obligations - Definition of Standards and Targets	HOD EMG
	7.2 Emissions Inventory, Calibration of equipment's	HOD(CHEMISTRY) / HOD(EMG)/ HOD(C&I)
	7.3 Assessment of Compliance	HOD EMG
	7.4 Need for plant Improvement	HOD EMG
	7.5 Improvement Planning	HOD(CHEMISTRY) / HOD(EMG)/ HOD(C&I)
	7.6 The Interface with the Regulatory Authority	HOD EMG
	7.7 Reporting	HOD(EMG)/ HOD(CHEMISTRY)

Annexure-B

Schedule for Monitoring

S.N	Activity	Location	Periodicity	Responsibility
2.1	Ambient Air Quality	1. Township 2. R&D Cabin 3. Kalpaka	Monthly	HOD(CHEMISTRY)
		4. Pittavanipalem 5. Dasaripeta 6. Devada. 7. Ravada	Fortnightly	
2.2	Ground Water Quality in Villages	Villages around ash pond	Monthly	HOD(CHEMISTRY)
2.3	CEMS		Continuous	HOD (C&I)/ HOD(O)/HOD (EMG)
2.4	Manual Stack monitoring	Stack 1 & 2	Fortnightly	HOD(CHEMISTR Y)/HOD (EMG)
2.5	Effluent Monitoring	1. STP Outlet 2. Common monitoring Basin	Three times a month	HOD(CHEMISTRY)
2.6	EQMS	EDP	Continuous	HOD (C&I)/ HOD (O)/HOD(EMG)

Annexure-C

DISTRIBUTION LIST

S.No.	TO
1.	HOP (Simhadri)
2.	HOD (O&M)
3.	HOD (MM)
4.	HOD (O)
5.	HOD (C&I)
6.	HOD (EM)
7.	HOD (EMG)
8.	HOD (CHEMISTRY)